

Networking Basics

Course Code: 3116

Course Title: Computer Networks



Dept. of Computer Science
Faculty of Science and Technology



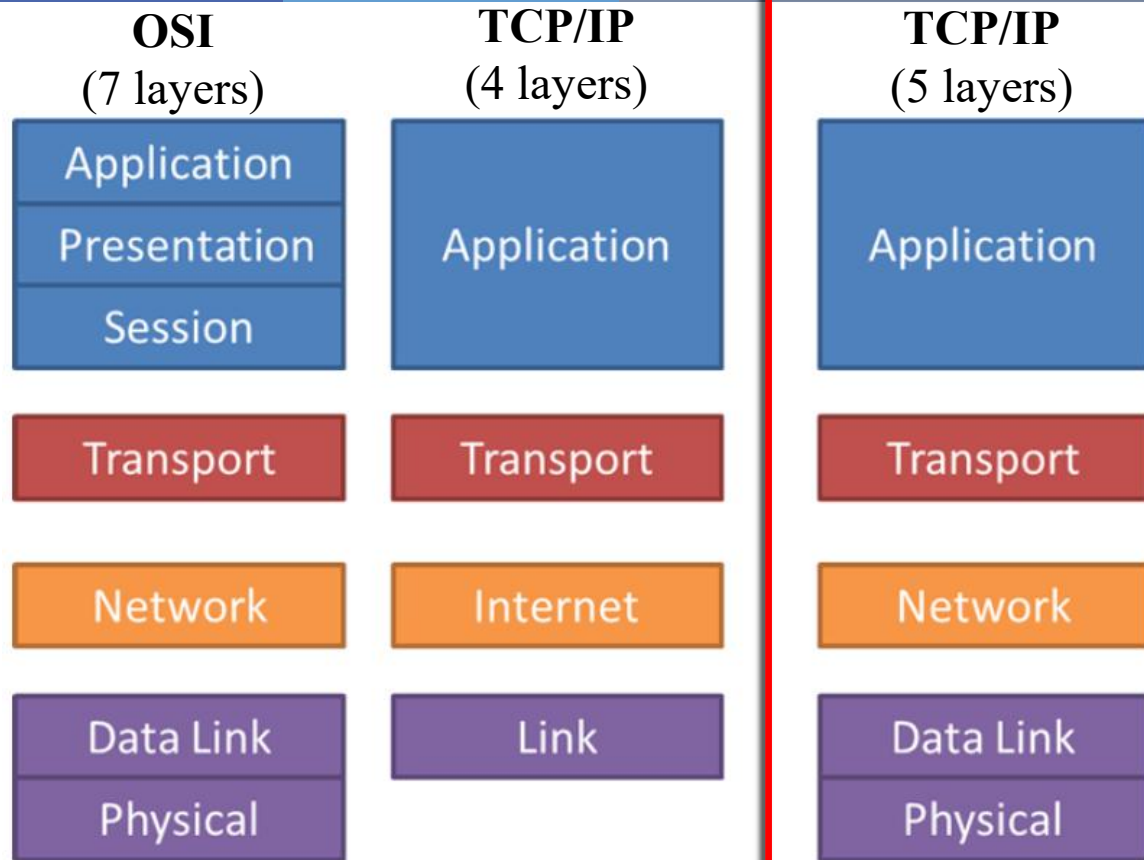
Lecturer No:	1	Week No:	1	Semester:	Fall 25-26
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Lecture Outline



1. Overview of TCP/IP Protocol Suite
2. Connecting Devices
 - Repeater
 - Hub
 - Bridge
 - Switch
 - Router
 - Modem
3. Collision domain and Broadcast domain
4. Ethernet Standards
5. WLAN Standards

OSI and TCP/IP protocol suites



OSI and TCP/IP protocol suites



	OSI	TCP/IP	Services/Protocols	Traffic
Software	Application	Application	HTTP, FTP, DNS, Telnet,	Data
	Presentation		ASCII, GIF, MPEG, ...	
	Session		Dialog control, Token Management, ..	
	Transport	Transport	TCP, UDP	Segments
	Network	Network	IP (IPv4 and IPv6), ICMP, IPsec, ...	Packets
Hardware	Data Link	Data Link	Ethernet, Switches, Frame Relay, ...	Frames
	Physical	Physical	0101, Hubs, Repeaters, ...	Bits

Overview of TCP/IP protocol suite



❖ Application Layer

- Only layer which interacts with **users applications**
- Takes data from users in sending end and provide the data to user in the receiving end.
- Protocols include HTTP, DNS, FTP, SMTP.
- Implemented in source and destination devices only

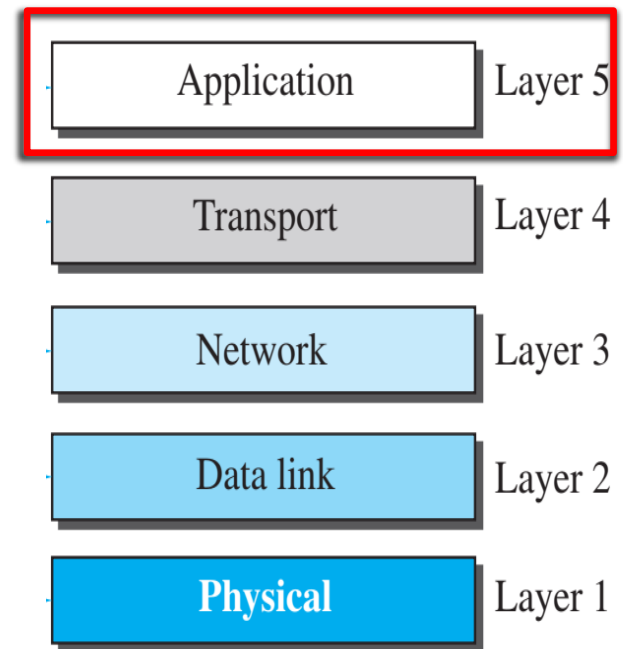


Fig. 1 Layers of TCP/IP protocol suites

Overview of TCP/IP protocol suite



❖ Transport Layer

- Gets the message from the application layer, encapsulates it in a segment and sends it to transport layer.
- Message delivery can be reliable but slow (TCP) or unreliable but fast (UDP)
- Provide port addressing to application layer programs
- Perform error control, flow control and congestion control

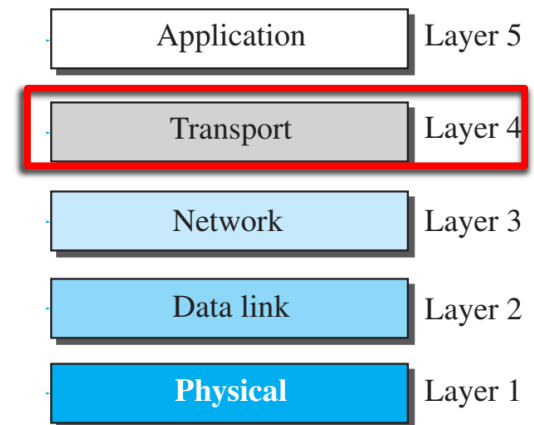


Fig. 1 Layers of TCP/IP protocol suites

Overview of TCP/IP protocol suite



❖ Transport Layer

- A **port address** identifies specific services or applications running on a device.
- Ports are represented by numbers (0-65535) and used by TCP/UDP protocols.
- Well-known ports:
 - 80 (HTTP - Web Traffic)
 - 443 (HTTPS - Secure Web Traffic)
 - 25 (SMTP - Email Sending)

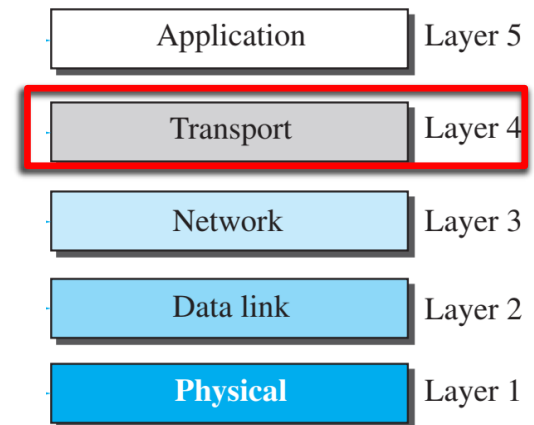


Fig. 1 Layers of TCP/IP protocol suites

Overview of TCP/IP protocol suite



❖ Network Layer or Internet Layer

- Require for communication between multiple networks
- Introduces IP address, perform routing and congestion control
- Encapsulate segment into a packet (called IP datagram)
- Used in end devices and network layer devices (Router, PC, Layer 3 switch, etc.)
- Protocols include IP ICMP, RIP, EIGRP, OSPF etc.

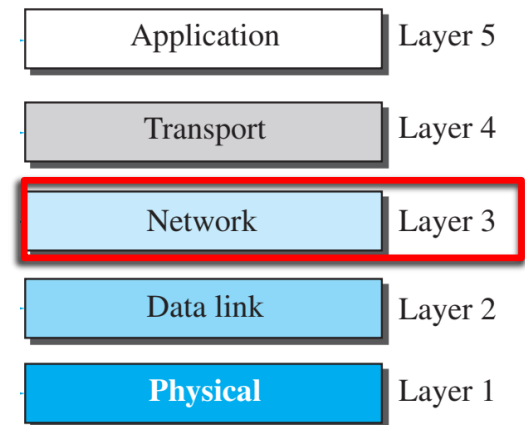


Fig. 1 Layers of TCP/IP protocol suites

Overview of TCP/IP protocol suite



❖ Network Layer or Internet Layer

- The **logical address** is the **IP** (Internet Protocol) **address**, assigned dynamically or statically.
- Identifying devices across different networks.
- Two types of IP addresses:
 - IPv4 (32-bit, e.g., 192.168.1.1)
 - IPv6 (128-bit, e.g., 2001:db8::ff00:42:8329)
- Routing packets between different networks.

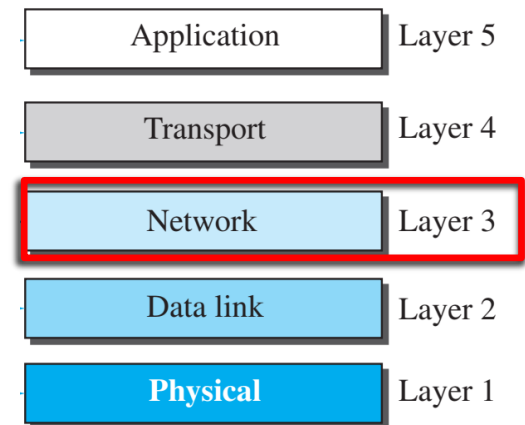


Fig. 1 Layers of TCP/IP protocol suites

Overview of TCP/IP protocol suite.



❖ Network Layer or Internet Layer

Classful addressing is the traditional method where IP addresses are divided into fixed classes (**A, B, C, D, and E**) based on predefined rules. Each class has a **fixed** subnet mask, and subnetting is not flexible.

Class	IP Range	Default Subnet Mask	Hosts per Network	Purpose
A	1.0.0.0 – 126.255.255.255	255.0.0.0 (/8)	16,777,214	Large Organizations
B	128.0.0.0 – 191.255.255.255	255.255.0.0 (/16)	65,534	Medium Organizations
C	192.0.0.0 – 223.255.255.255	255.255.255.0 (/24)	254	Small Networks
D	224.0.0.0 – 239.255.255.255	N/A	N/A	Multicasting
E	240.0.0.0 – 255.255.255.255	N/A	N/A	Research & Reserved

Overview of TCP/IP protocol suite



❖ Data-link Layer

- Require for communication inside a network
- Introduces MAC address, perform error control and flow control
- Encapsulating packet into frame
- Used in all devices (PC, Router, Switch, Bridge) except hub & repeater
- Protocols include ALOHA, CSMA, CSMA/CD, CSMA/CA

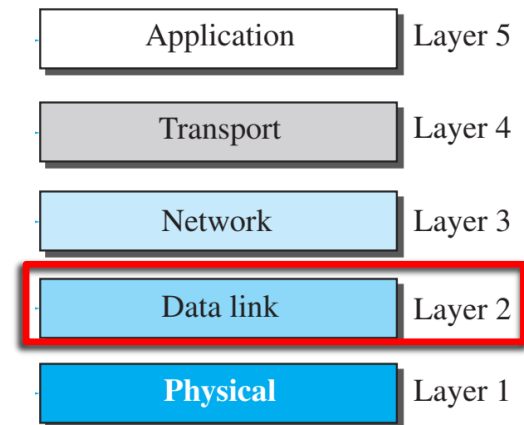


Fig. 1 Layers of TCP/IP protocol suites

Overview of TCP/IP protocol suite



❖ Data-link Layer

- The **physical address** is also known as the **MAC (Media Access Control) address**. It is a **unique** identifier assigned to a **network interface card (NIC)** by the manufacturer.
- MAC addresses are **48 bits long** and written in **hexadecimal format** (e.g., 00:A1:B2:C3:D4:E5).
- Consider two computers on a LAN. Each has a unique MAC address:
PC1: 00:1A:2B:3C:4D:5E
PC2: 00:1A:2B:3C:4D:5F

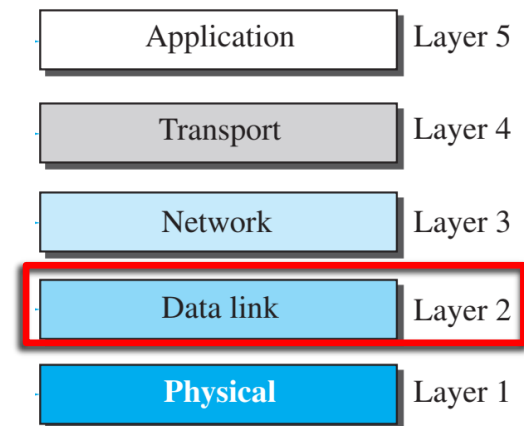


Fig. 1 Layers of TCP/IP protocol suites

Overview of TCP/IP protocol suite



❖ Physical Layer

- Accepts a complete frame from the Data Link layer and encodes it as a series of signals that are transmitted onto the local media.
- Specifies transmission media
- Network physical topology

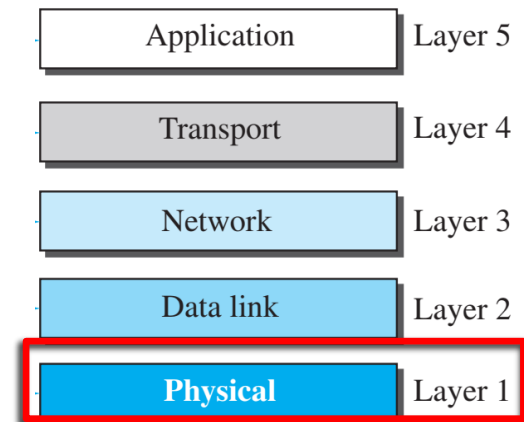
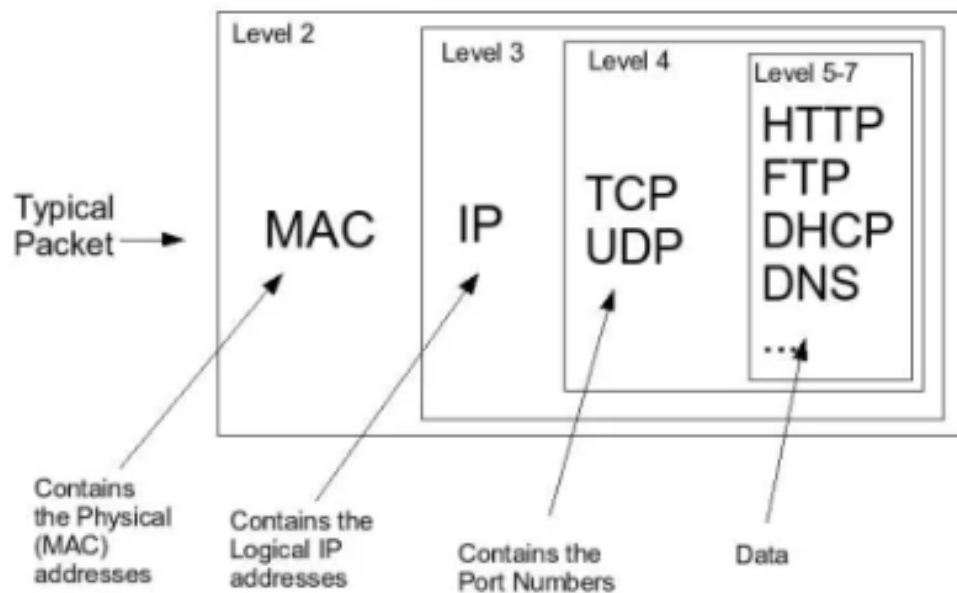


Fig. 1 Layers of TCP/IP protocol suites



protocol suite...



Legend

- 4 Header at transport layer
 - 3 Header at network layer
 - 2 Header at data-link layer
- ↓ Encapsulate
- ↑ Decapsulate

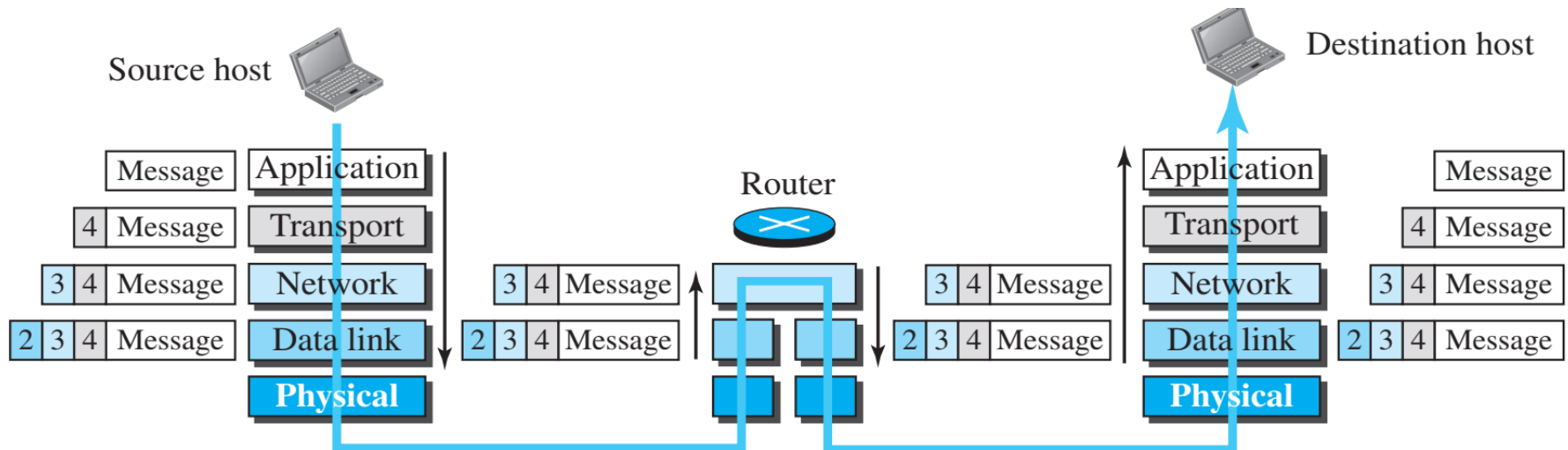


Fig. 2 Encapsulation/Decapsulation

Connecting Devices



Switch



Router



Bridge



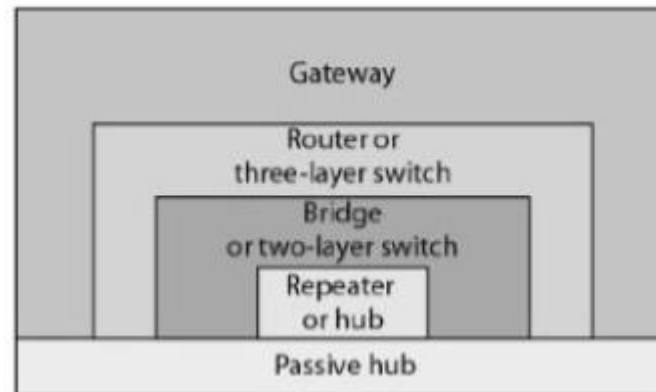
Repeater



Hub



Modem



Application
Transport
Network
Data link
Physical

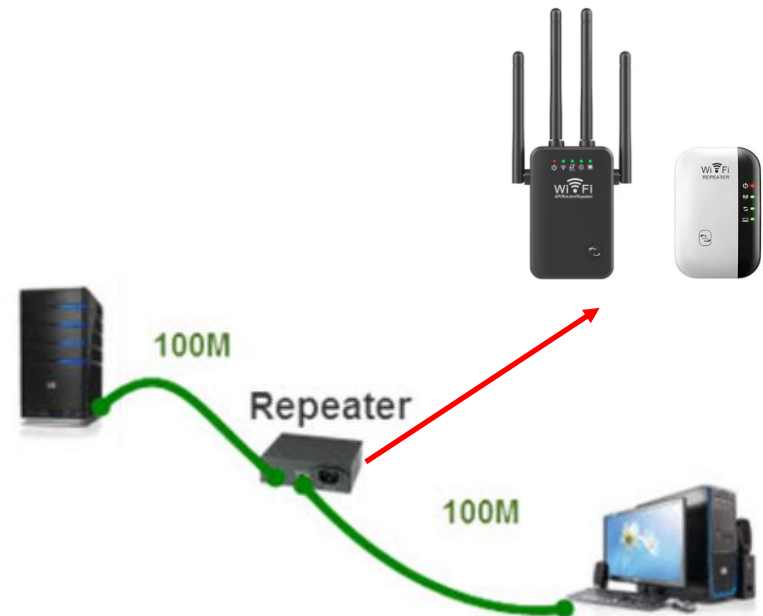
Connecting Devices

Repeater



❖ Repeater

- **Layer 1 device** that takes voltage from the line, amplifies the voltage, and sends it down the line.
- Used to **extend a network** beyond the maximum length of the cable segment
- If there is any “noise” caused by electromagnetic interference on the wire, it will also amplify the noise and send it.
- **Using 3 repeaters in a row results in unusable signal transmission due to excessive noise.**
- These devices are not in common use anymore; they have been replaced by hubs, bridges, and switches.



Connecting Devices....

Repeater

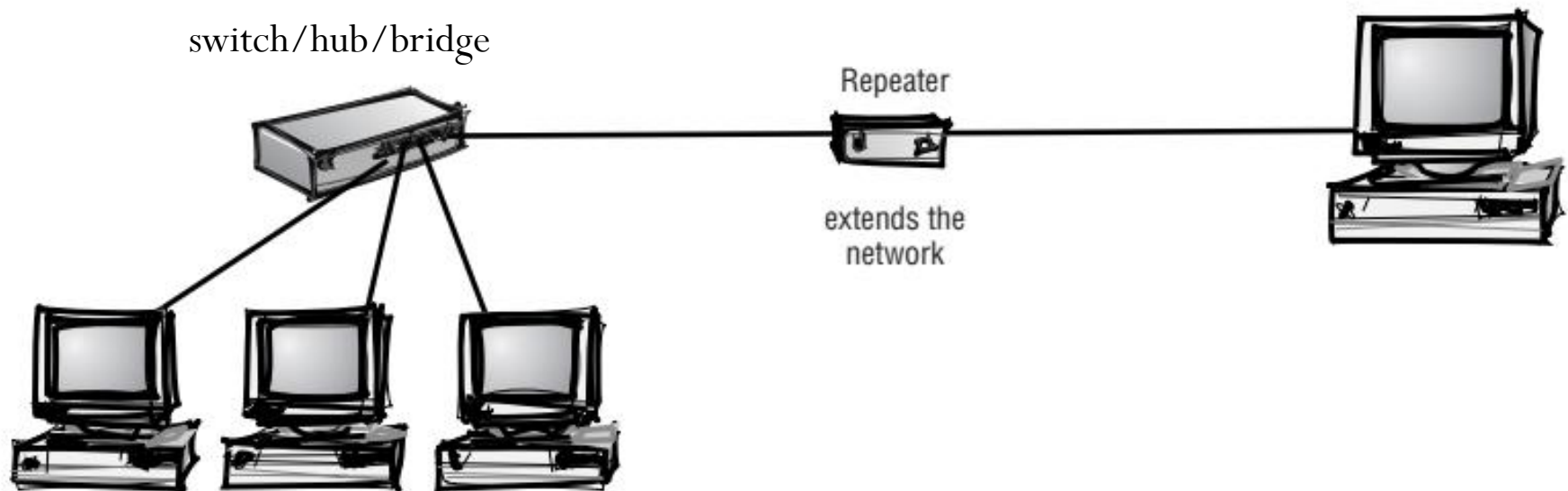


Fig. 3 Use of repeaters and switch/hub/bridge [2]

Connecting Devices....

Hub



❖ Hub

- A hub with more than one output port.
- Electrical signal comes through one port of the hub and gets amplified and sent out through all ports of the hub.
- If you have a **10-Mbps hub** and **three devices** are transmitting at the same time, each device gets **one third of the bandwidth**.
- For a successful transmission, only one station can send data at a time.
- More active ports cause more collision among signal, thereby resulting in lower data rate.
- A **layer 1 device** and is Used to connect devices of a single network

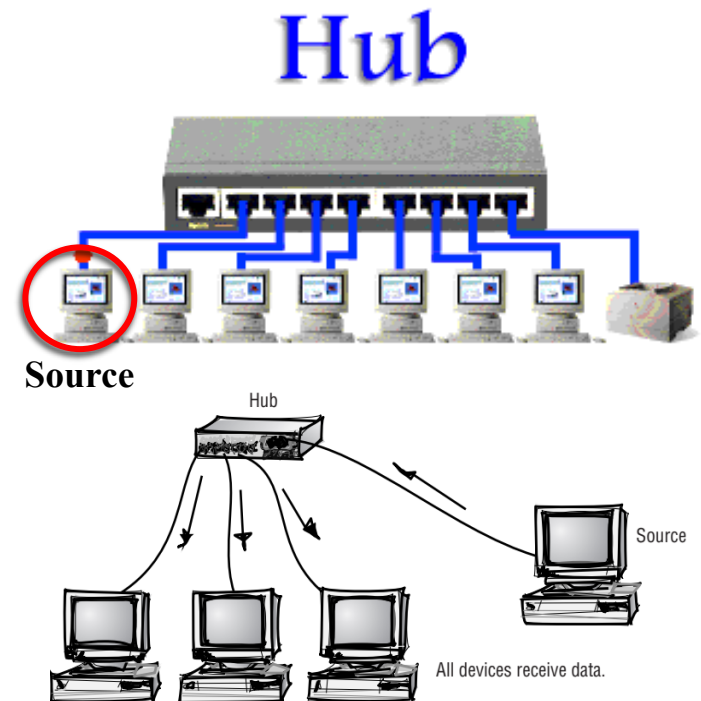


Fig. 4 Broadcasting of a Hub

Connecting Devices....

Bridge



❖ Bridge

- A **layer 2** device
- Used to connect devices of a single network
- It sends the received frame only to the intended destination based on the destination MAC address of the frame.
- Better bandwidth usage.
- Unlike hub, it has error detection capability
- Limited ports (2-4, usually 2 ports)

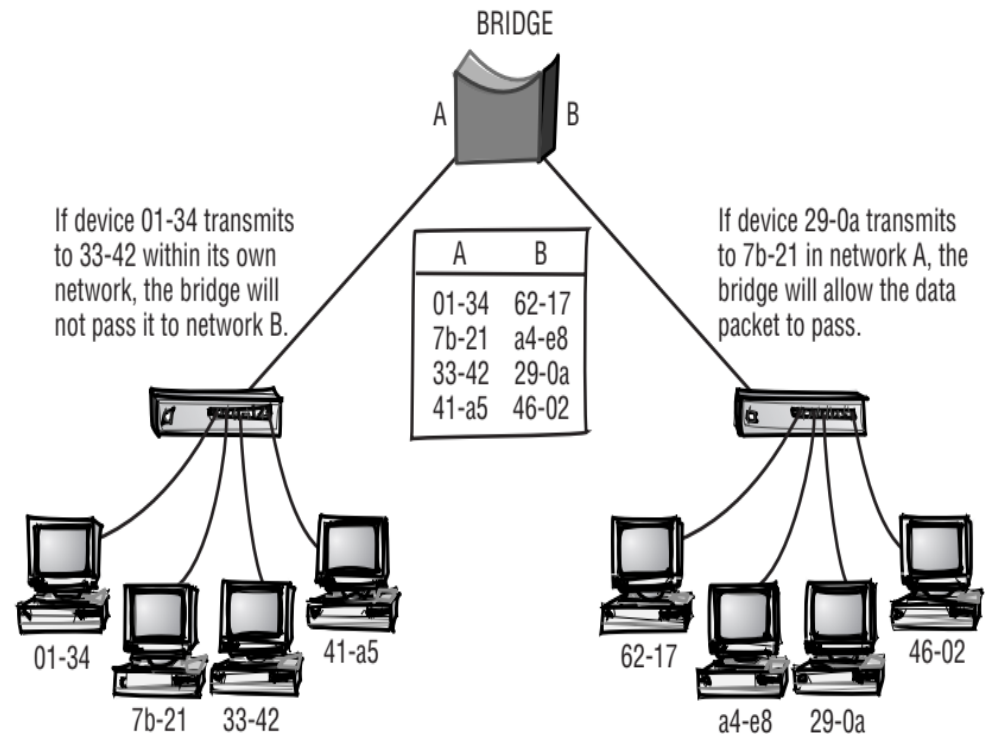


Fig. 5 Filtering of a Bridge

Connecting Device

Switch

❖ Switch

- A **layer 2** device
- Used to **connect devices of a single network**
- **Like a Bridge**, it sends the received **frame** only to the intended destination based on the destination MAC address of the frame.
- Unlike hub, it has error detection capability
- Hundreds of ports (2 to more than 100 ports)
- Frame forwarding decision is taken based on hardware, hence **it is faster than bridge**.
- If you have a 10-Mbps switch with three devices connected to it, all **three devices can use 10-Mbps of bandwidth**.

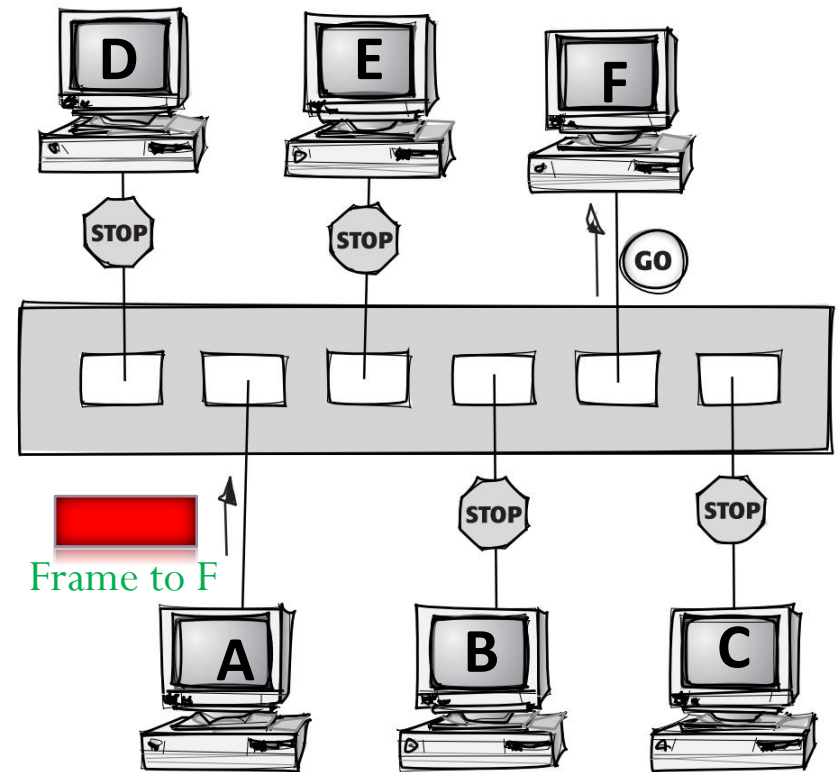


Fig. 6 Filtering of a Switch

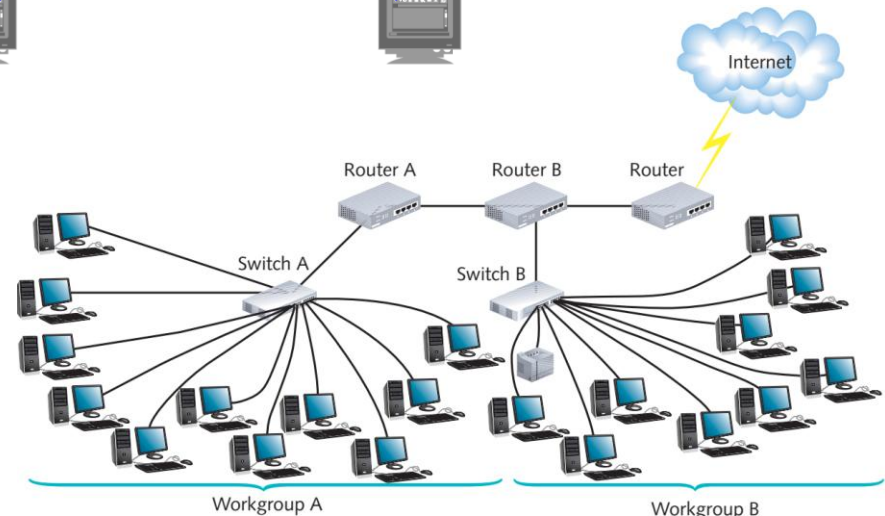
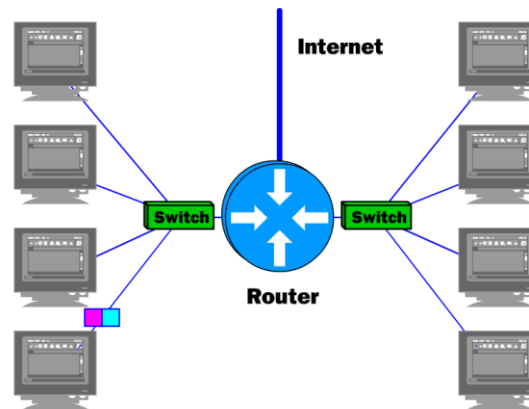
Connecting Devices....

Router



❖ Router

- A **layer 3 device**
- Used to **connect multiple networks**
- Connected networks can have **different protocols and speed.**
- Forward packets based on destination IP address
- **Most intelligent connecting device**
- Can also be used to forward packet within a network
- **Slower than switch** because of its routing protocol implementation



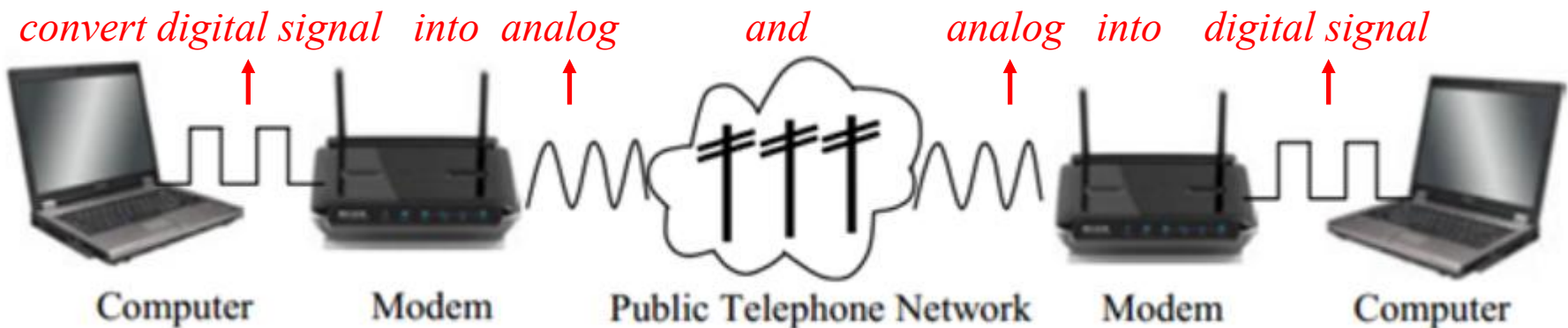
Connecting Devices....

Modem



❖ Modem

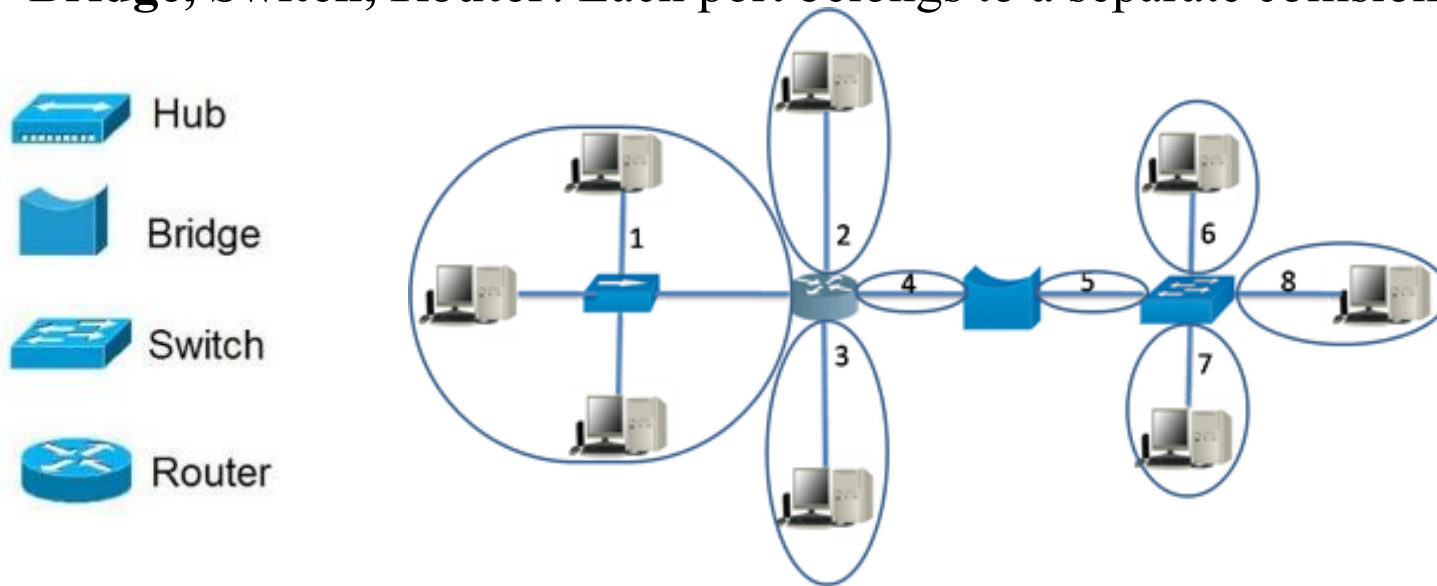
- A modem (short for modulator-demodulator) is a device that enables digital data to be transmitted over analog communication lines, such as telephone lines or cable systems.
- The **modulator** converts digital data into analog data when the data is being sent by the computer.
- The **demodulator** converts analog data signals into digital data when it is being received by the computer.



Collision Domain



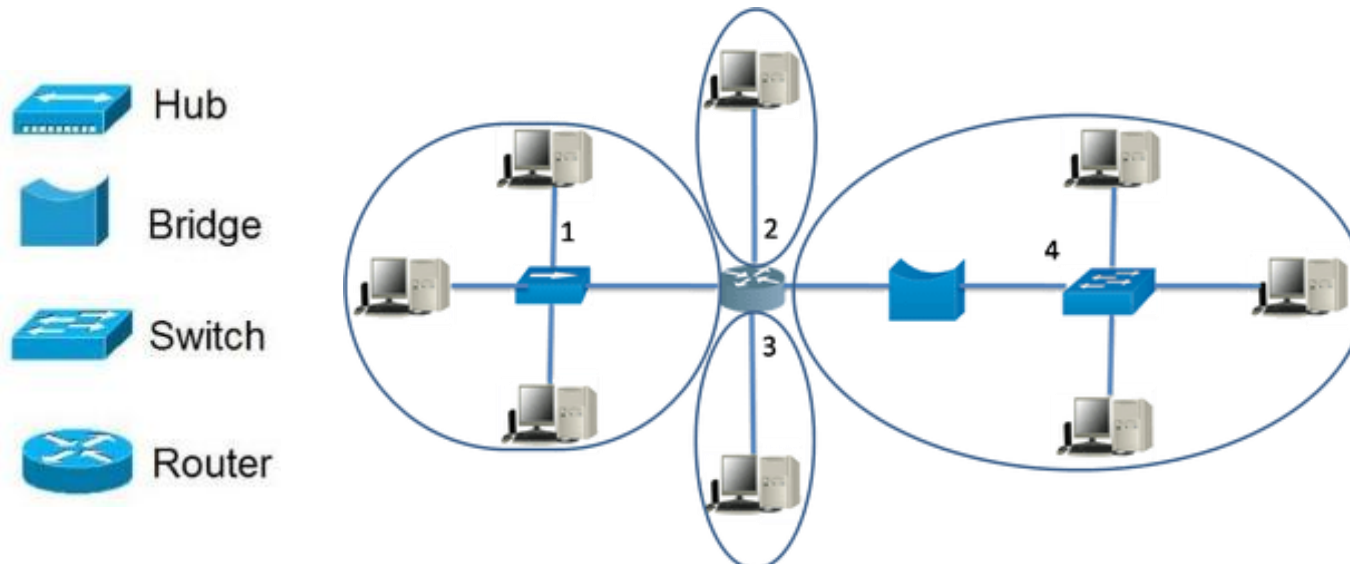
- The “collision domain” describes a network where packet collisions can occur when two devices on a **shared network medium** send packets simultaneously.
- **Hub**: All ports belong to the same collision domain.
- **Bridge, Switch, Router**: Each port belongs to a separate collision domain.



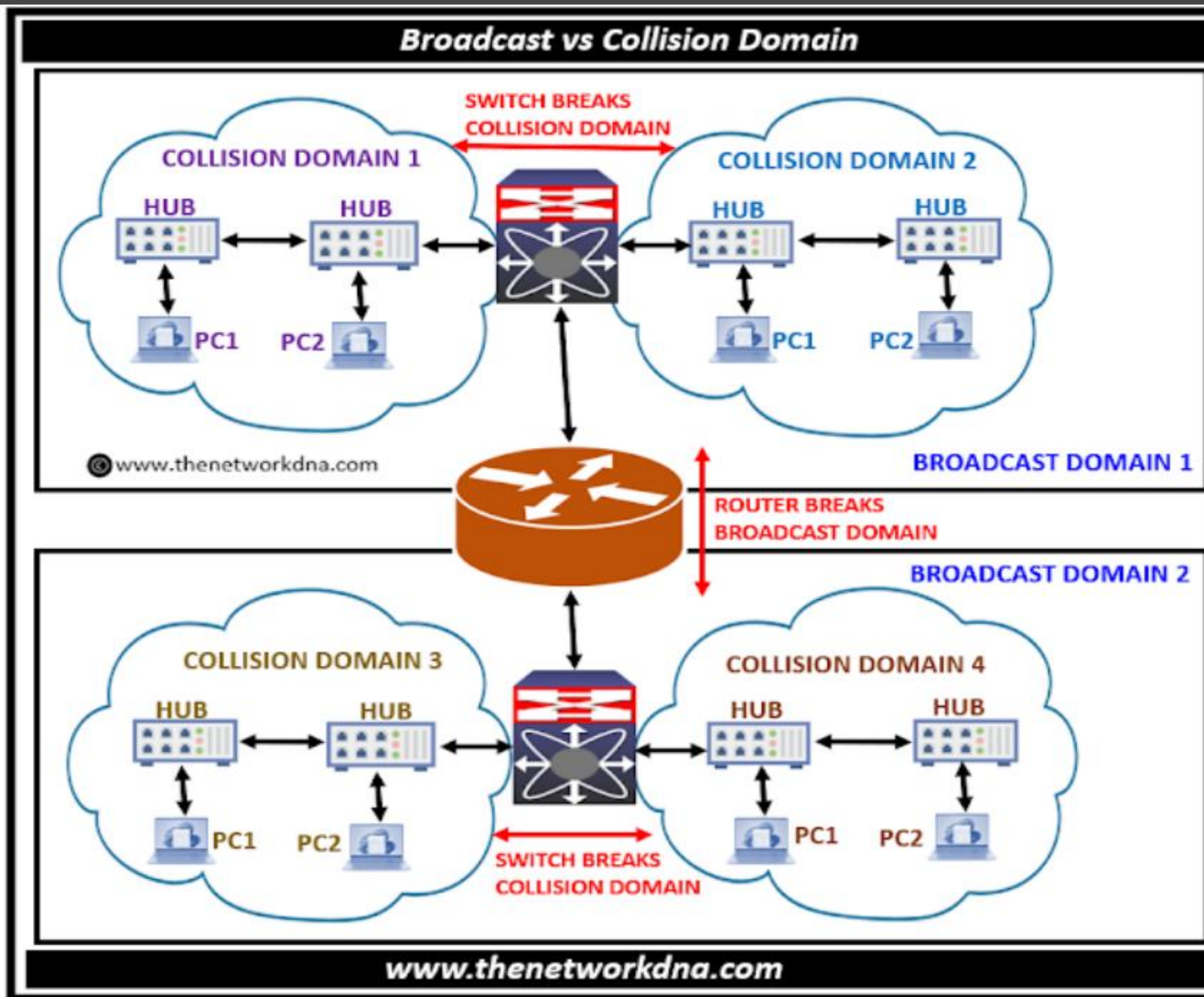
Broadcast Domain



- All the devices in the broadcast domain can reach via broadcast at the data link layer.
- A **Broadcast Domain** can receive any broadcast packet originating from any device within the network segment.
- All ports of hub and switch belong to same broadcast domain but all ports of the router belong to different broadcast domain.

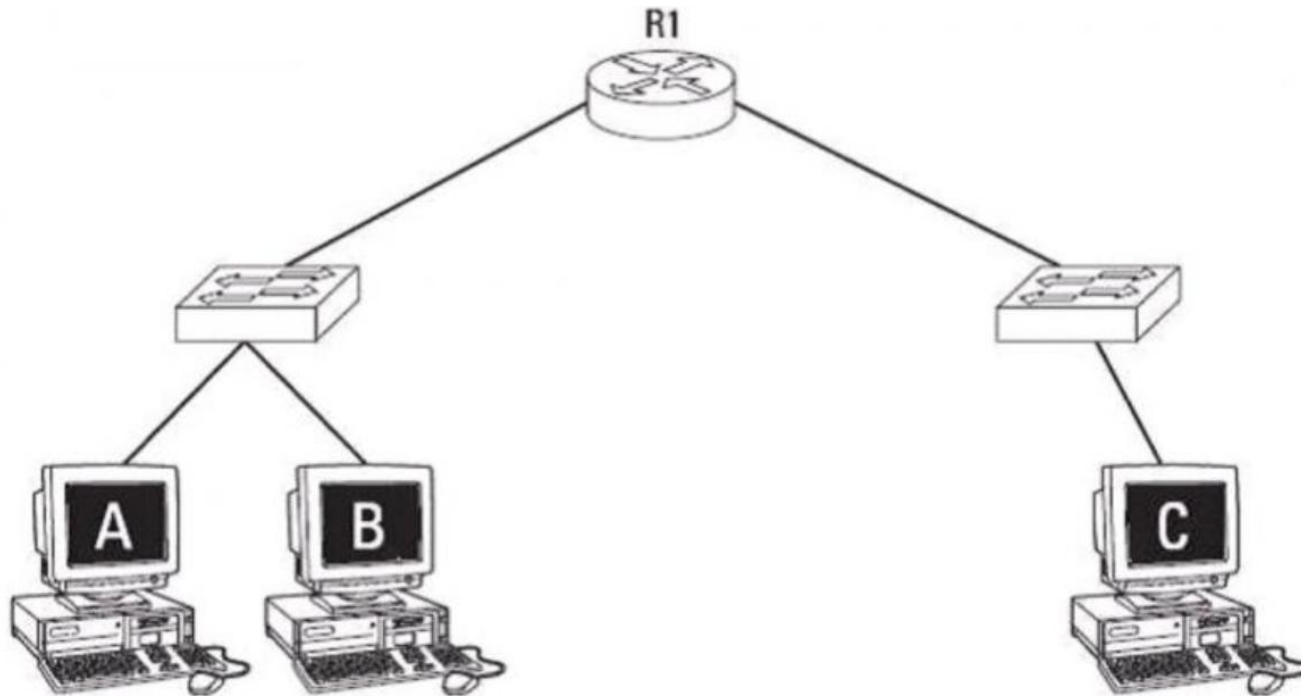


Broadcast vs Collision Domain



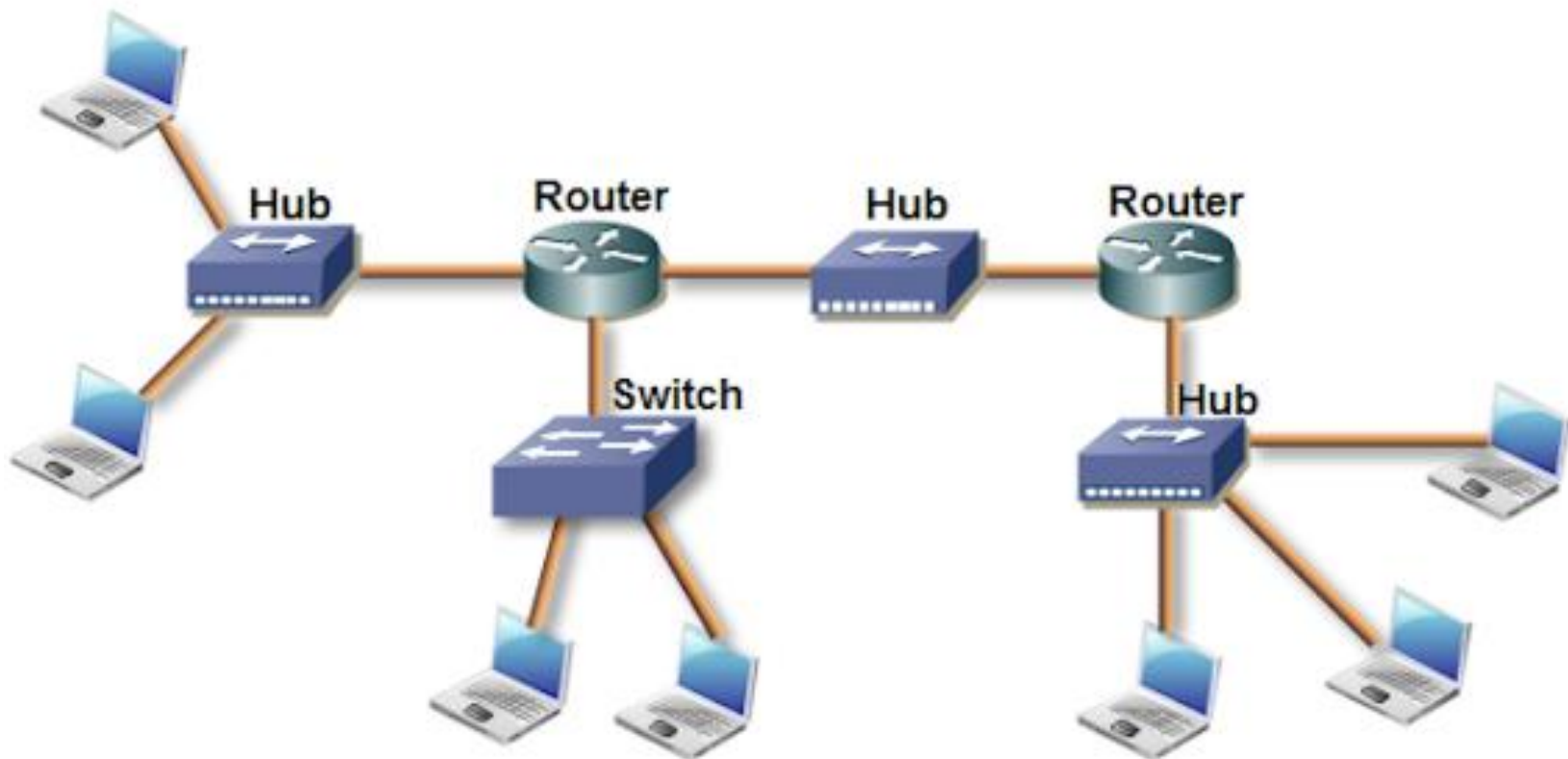
Broadcast vs Collision Domain

How many collision and broadcast domains are there in the following network diagram?



Task

Find collision and broadcast domain for the following network:



Ethernet Standards



TABLE I Ethernet Standard

Speed	Common Name	Informal Standard Name	Formal Standard Name	Cable Type	Max. Length
10 Mbps	Ethernet	10BASE-T	802.3	Cat3	100 m
100 Mbps	Fast Ethernet	100BASE-T	802.3u	Cat5	100 m
1000 Mbps	Gigabit Ethernet	1000BASE-LX	802.3z	Single mode fiber	5000 m
				50-micron multimode fiber	550 m
				62.5-micron multimode fiber	440 m
1000 Mbps	Gigabit Ethernet	1000BASE-T	802.3ab	Cat5, Cat5e	100 m
10 Gbps	10 Gig bit Ethernet	10GBASE-T	802.3an	Cat6, Cat6a	100 m

WLAN Standards



TABLE II WLAN Standard

Release date	Standard	Frequency band	Bandwidth	Transmission scheme	Max modulation	MIMO	Max data rate
1997	802.11	2.4 GHz	20 MHz	DSSS, FHSS	QPSK	N/A	2 Mbps
1999	802.11b	2.4 GHz	20 MHz	DSSS	QPSK	N/A	11 Mbps
1999	802.11a	5 GHz	20 MHz	OFDM	64 QAM	N/A	54 Mbps
2003	802.11g	2.4 GHz	20 MHz	DSSS, OFDM	64 QAM	N/A	54 Mbps
2009	802.11n	2.4 GHz 5 GHz	20 MHz 40 MHz	OFDM	64 QAM	4×4	600 Mbps
2013	802.11ac	5 GHz	20 MHz 40 MHz 80 MHz 160 MHz	OFDM	256 QAM	8×8	6.93 Gbps
2018	802.11ad	60 GHz	2160 MHz	SC-FDM, OFDM	256 QAM	Beamforming	6.93 Mbps



Recommended Books

1. **Data Communications and Networking**, *B. A. Forouzan*, McGraw-Hill, Inc., Fourth Edition, 2007, USA.
2. **Computer Networking: A Top-Down Approach**, *J. F. Kurose, K. W. Ross*, Pearson Education, Inc., Sixth Edition, USA.
3. **Official Cert Guide CCNA 200-301 , vol. 1**, *W. Odom*, Cisco Press, First Edition, 2019, USA.
4. **CCNA Routing and Switching**, *T. Lammle*, John Wiley & Sons, Second Edition, 2016, USA.
5. **TCP/IP Protocol Suite**, *B. A. Forouzan*, McGraw-Hill, Inc., Fourth Edition, 2009, USA.
6. **Data and Computer Communication**, *W. Stallings*, Pearson Education, Inc., Tenth Edition, 2013, USA.